

Bayesian Age-Period-Cohort Modeling

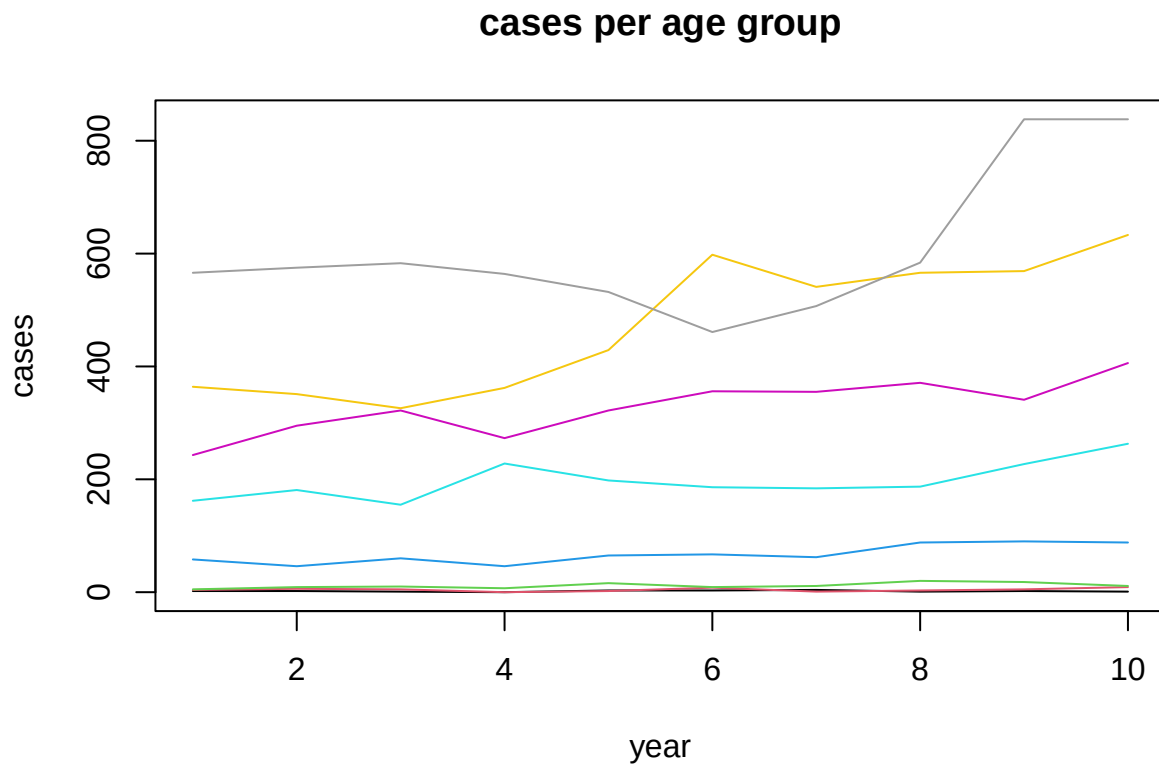
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Data example

BAMP includes a data example.

```
data(apc)
plot(cases[,1],type="l",ylim=range(cases), ylab="cases", xlab="year", main="cases per age group")
for (i in 2:8)lines(cases[,i], col=i)
```



APC model with random walk first order prior

```
model1 <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
              periods_per_agegroup = 5)
```

bamp() automatically performs a check for MCMC convergence using Gelman and Rubin's convergence diagnostic. We can manually check the convergence again:

```
checkConvergence(model1)
```

```
## [1] TRUE
```

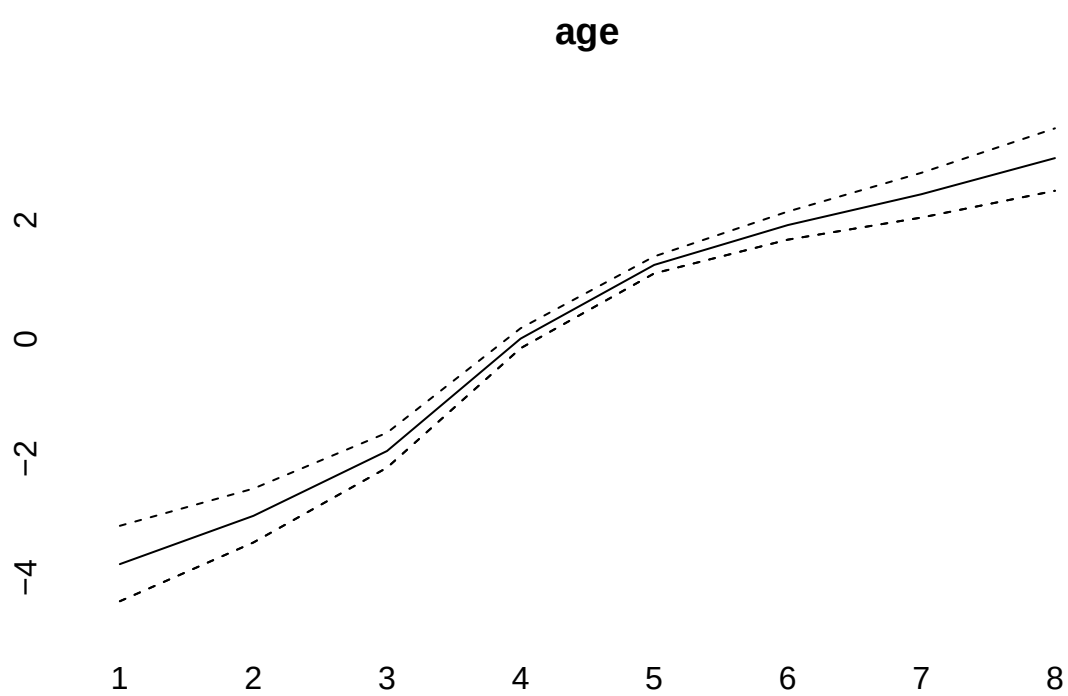
Now we have a look at the model results. This includes estimates of smoothing parameters and deviance and DIC:

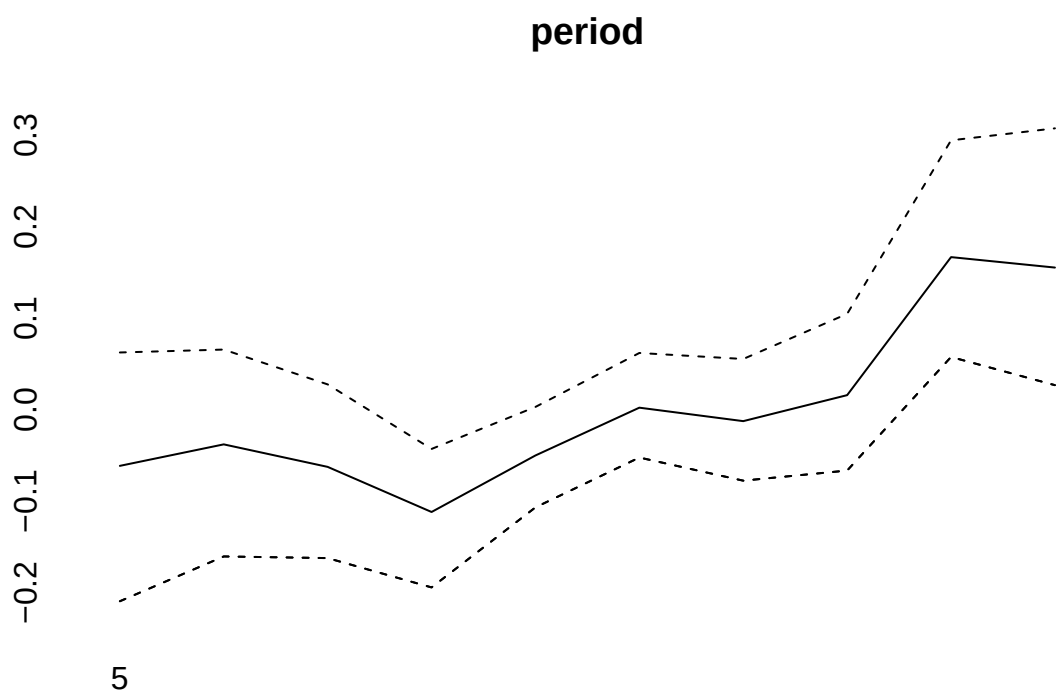
```
print(model1)
```

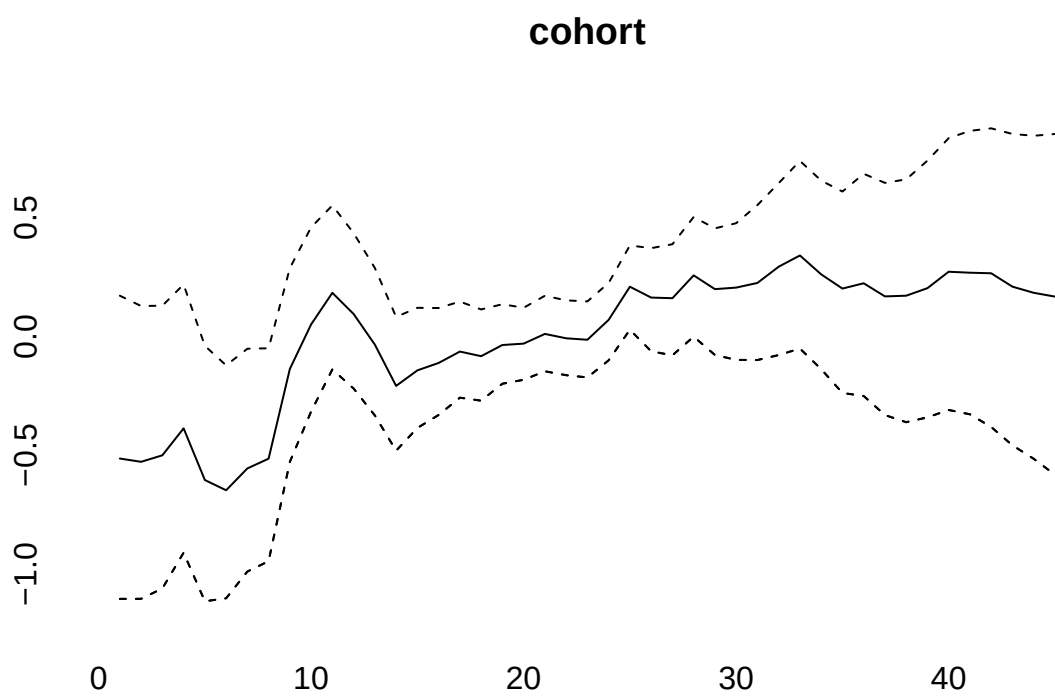
```
##
## Model:
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      231.15
## pD:            36.91
## DIC:           268.06
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.369         0.918         1.961
## period                     68.621        200.175        621.471
## cohort                     34.187         59.267         98.439
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

We can plot the main APC effects using point-wise quantiles:

```
plot(model1)
```

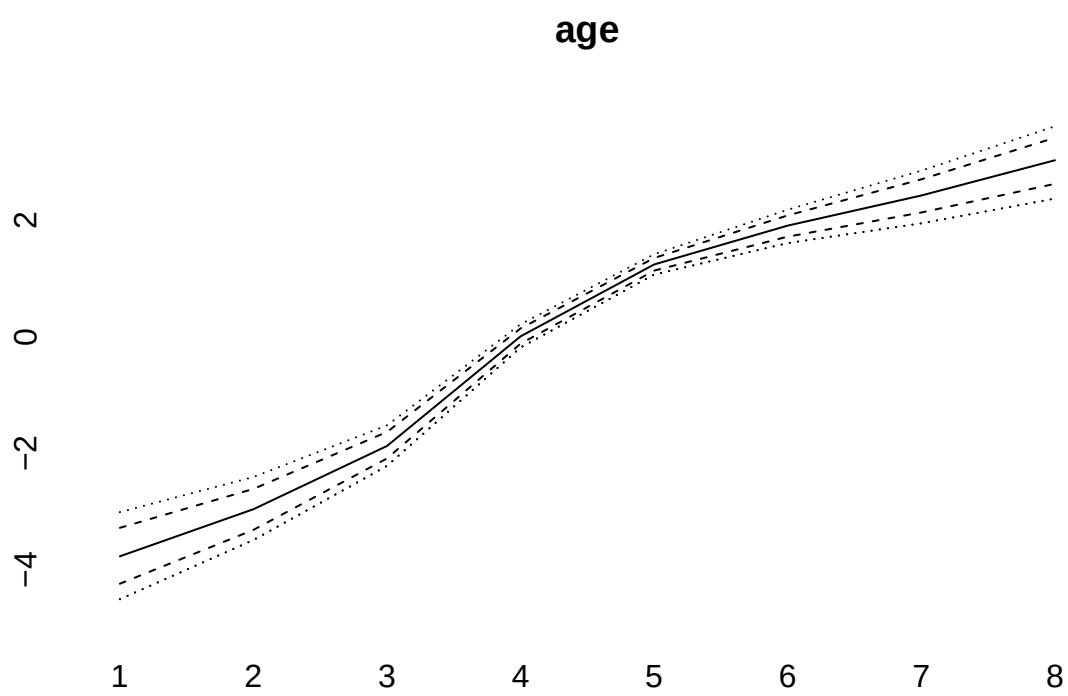


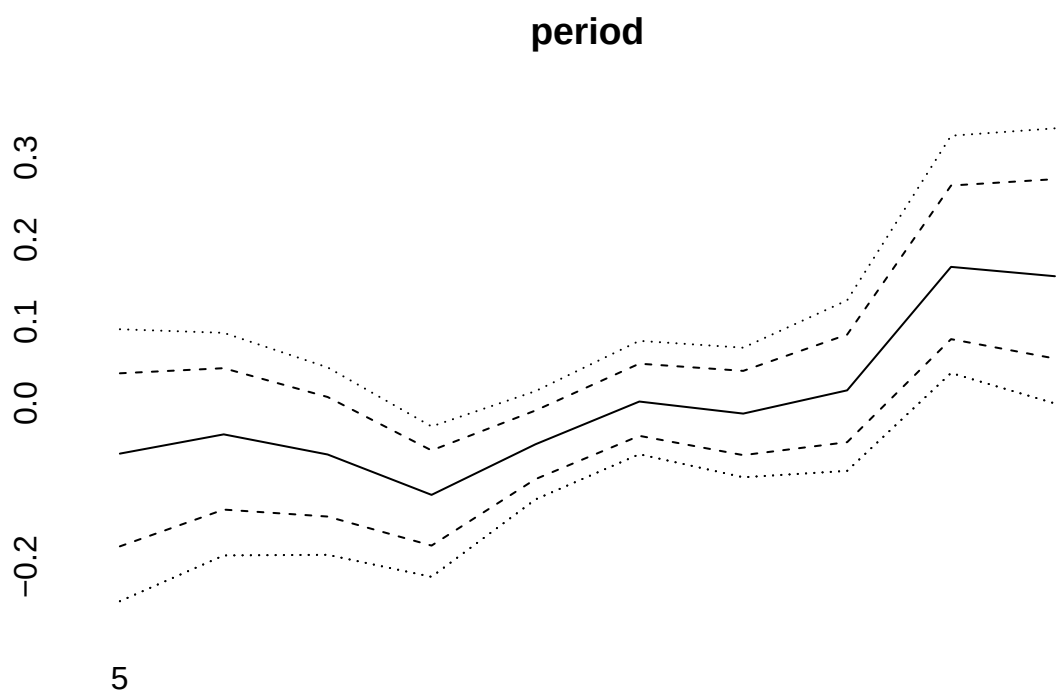


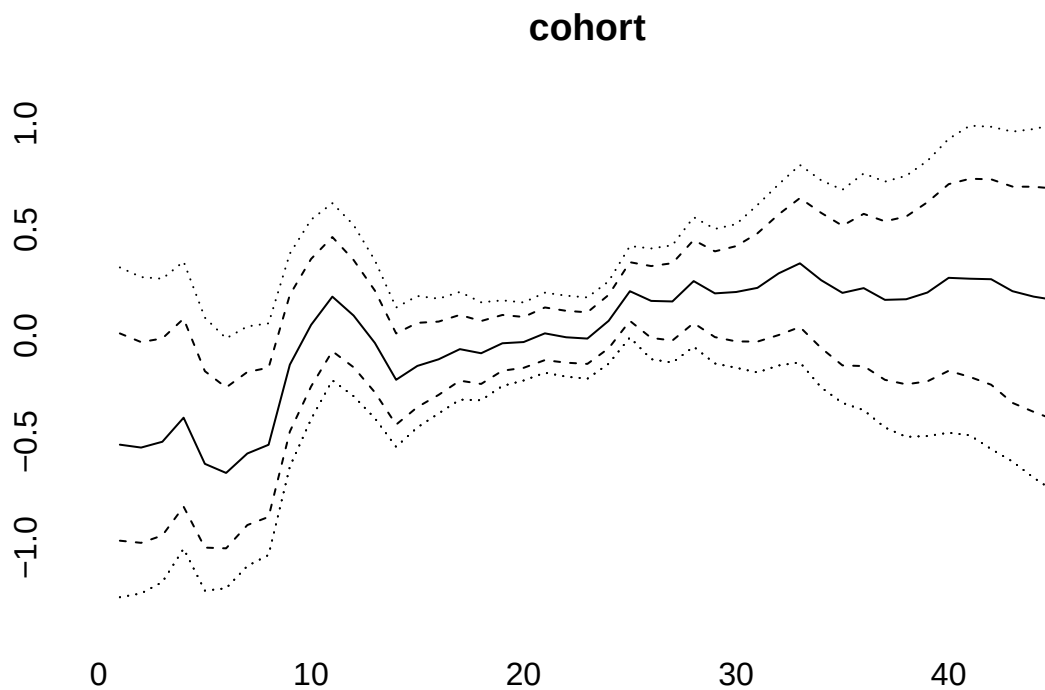


More quantiles are possible:

```
plot(model1, quantiles = c(0.025,0.1,0.5,0.9,0.975))
```







APC model with random walk second order prior

```
model2 <- bamp(cases, population, age="rw2", period="rw2", cohort="rw2",
               periods_per_agegroup = 5,
               mcmc.options=list("number_of_iterations"=200000, "burn_in"=100000, "step"=50, "tuning"=50),
               hyperpar=list("age"=c(1,.5), "period"=c(1,0.05), "cohort"=c(1,0.05)))
```

```
checkConvergence(model2)
```

```
## [1] TRUE
```

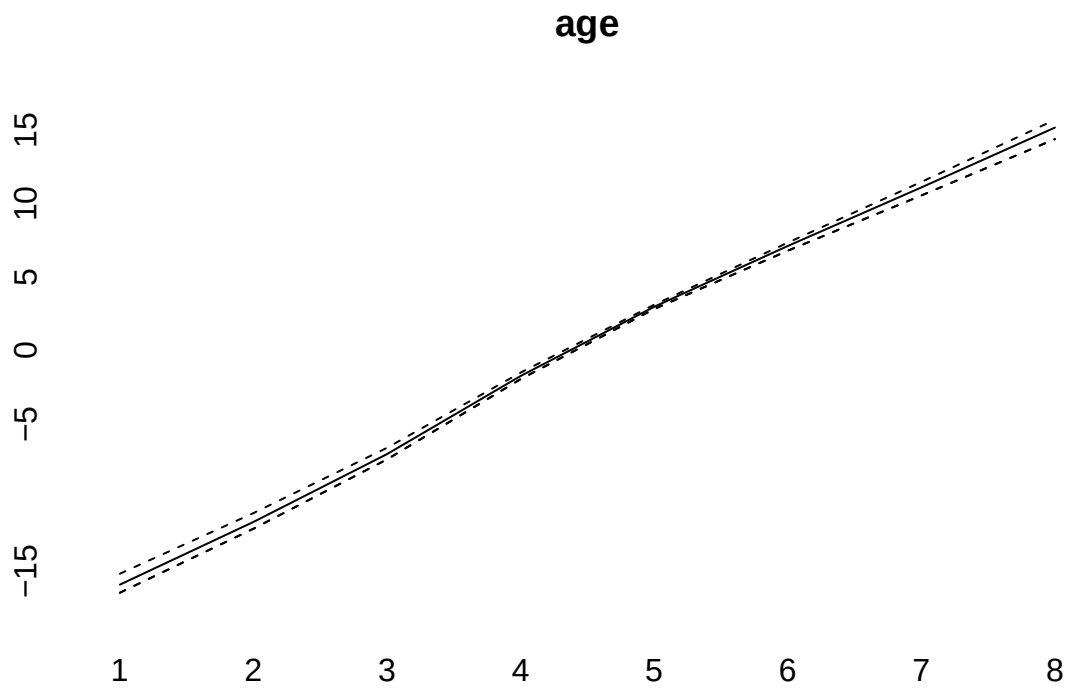
```
print(model2)
```

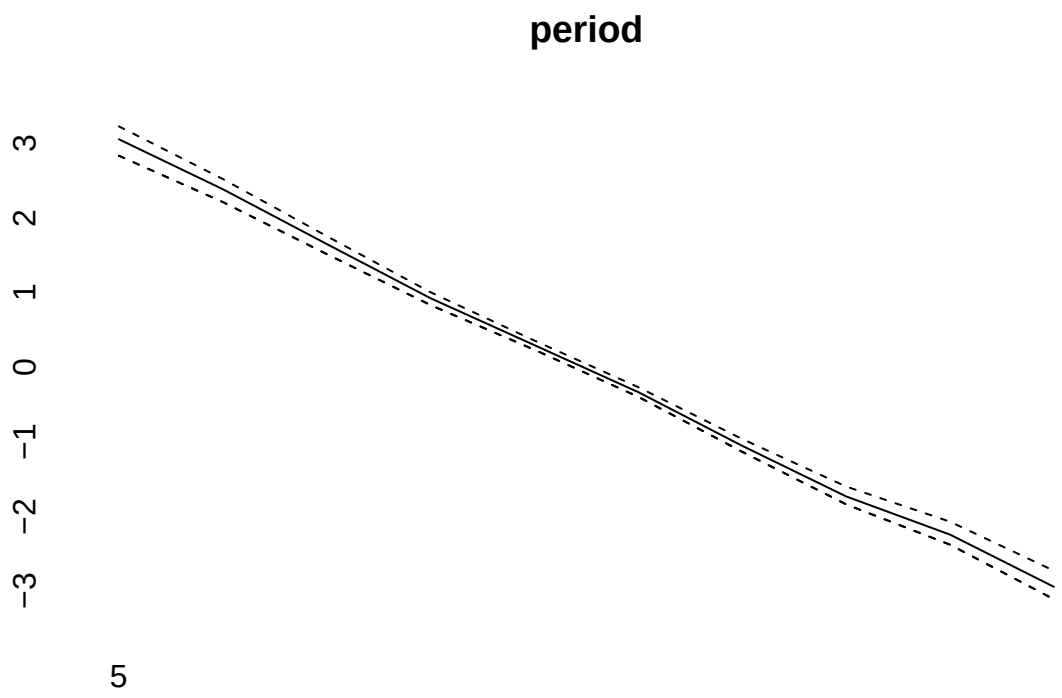
```
##
## Model:
## age (rw2) - period (rw2) - cohort (rw2) model
## Deviance:      234.42
## pD:            36.99
## DIC:           271.41
##
##
## Hyper parameters:           5%           50%           95%
```

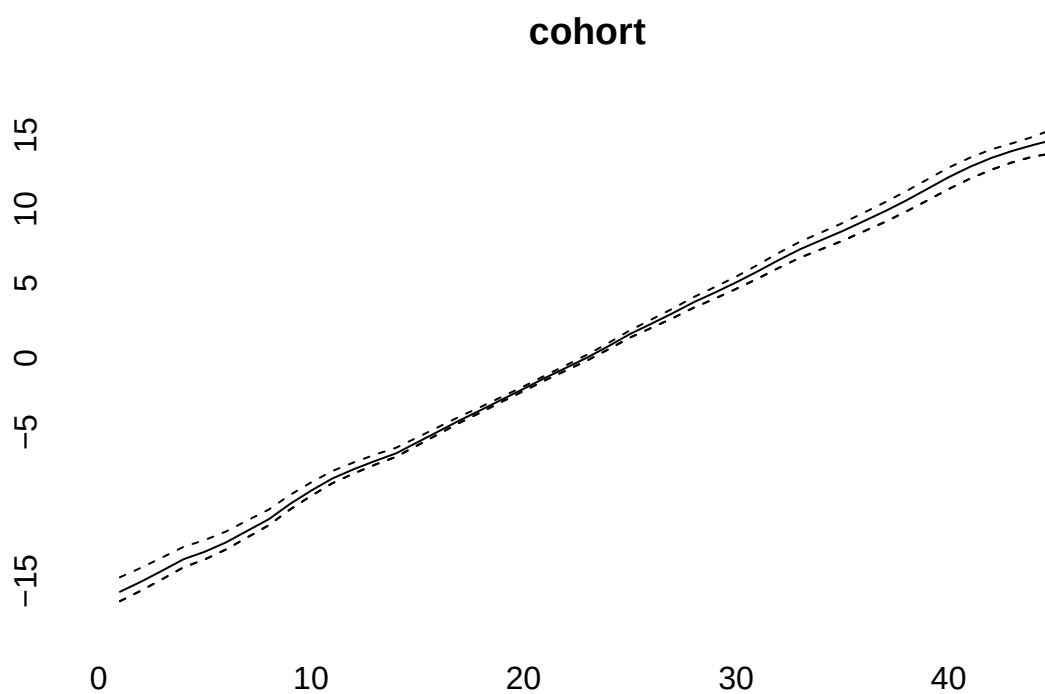


```
## age          1.035      2.899      6.652
## period      16.060     41.574     90.199
## cohort      23.287     45.096     82.014
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model2)
```







```
model3<-bamp(cases, population, age="rw1", period=" ", cohort="rw2",
             periods_per_agegroup = 5)
```

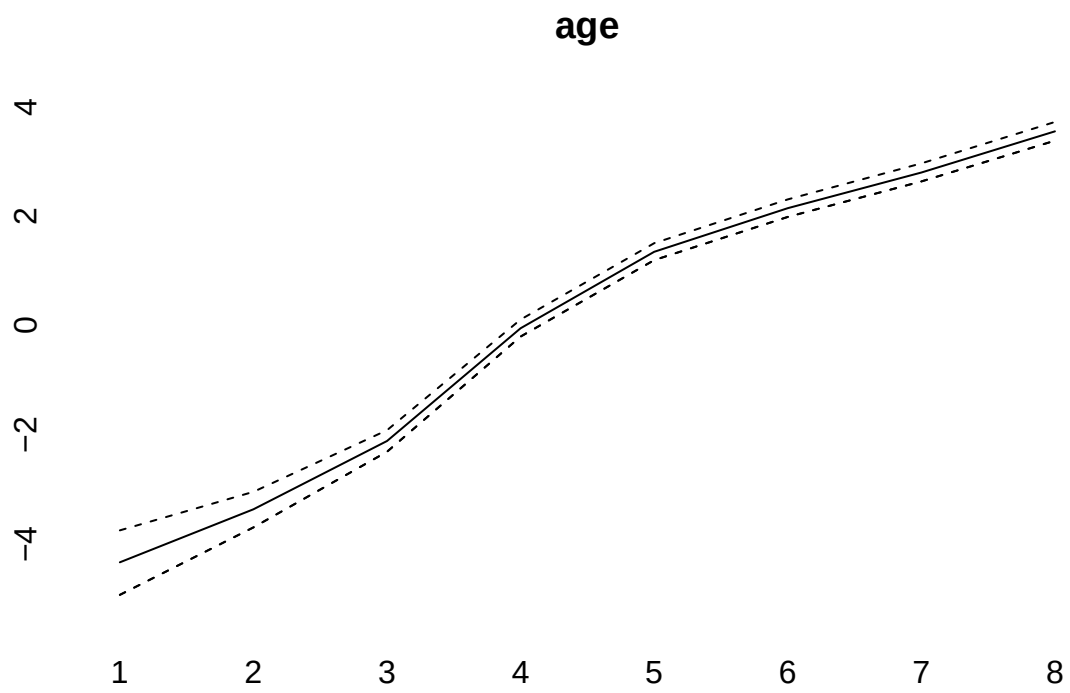
```
checkConvergence(model3)
```

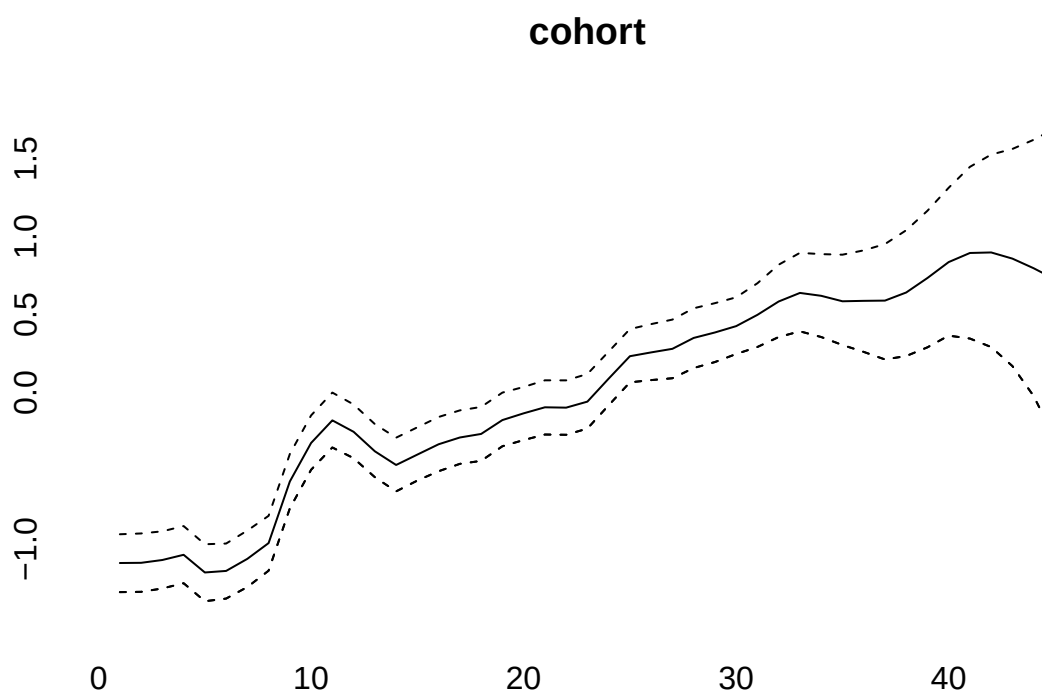
```
## [1] TRUE
```

```
print(model3)
```

```
##
## Model:
## age (rw1) cohort (rw2) model
## Deviance:      276.72
## pD:            30.17
## DIC:           306.89
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.294        0.724        1.491
## cohort                     39.329       74.561      139.182
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model3)
```

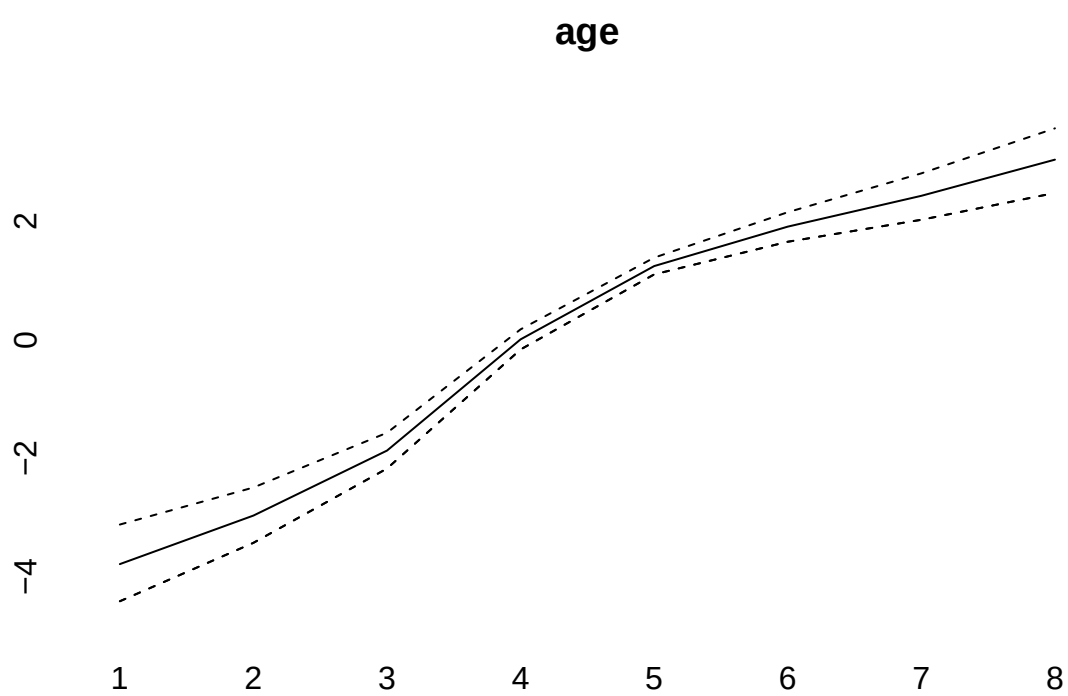


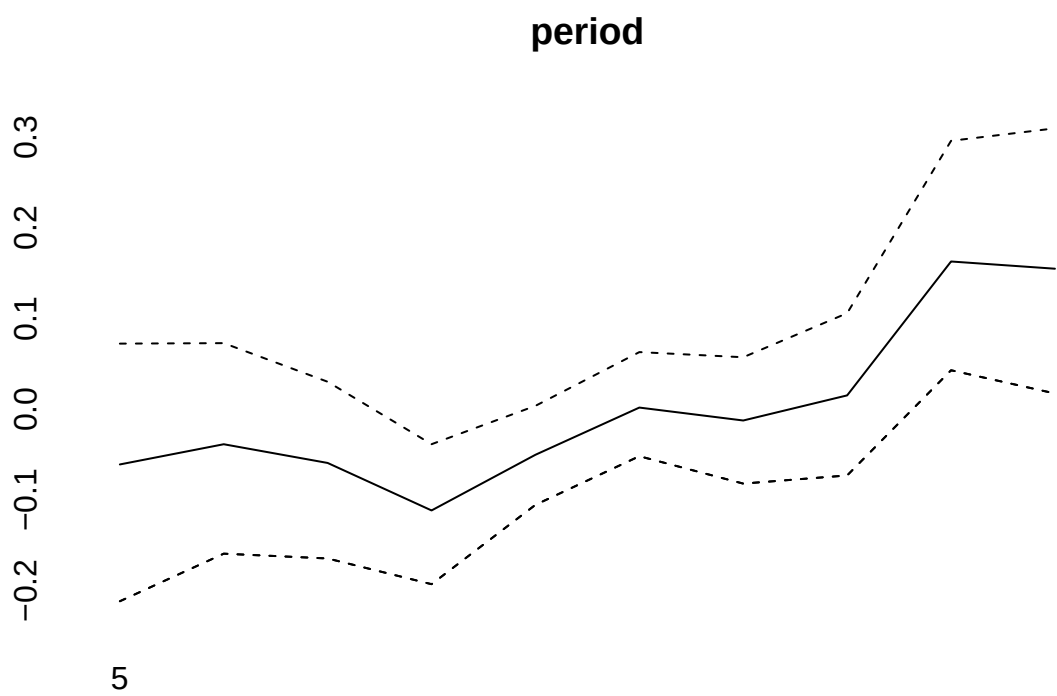


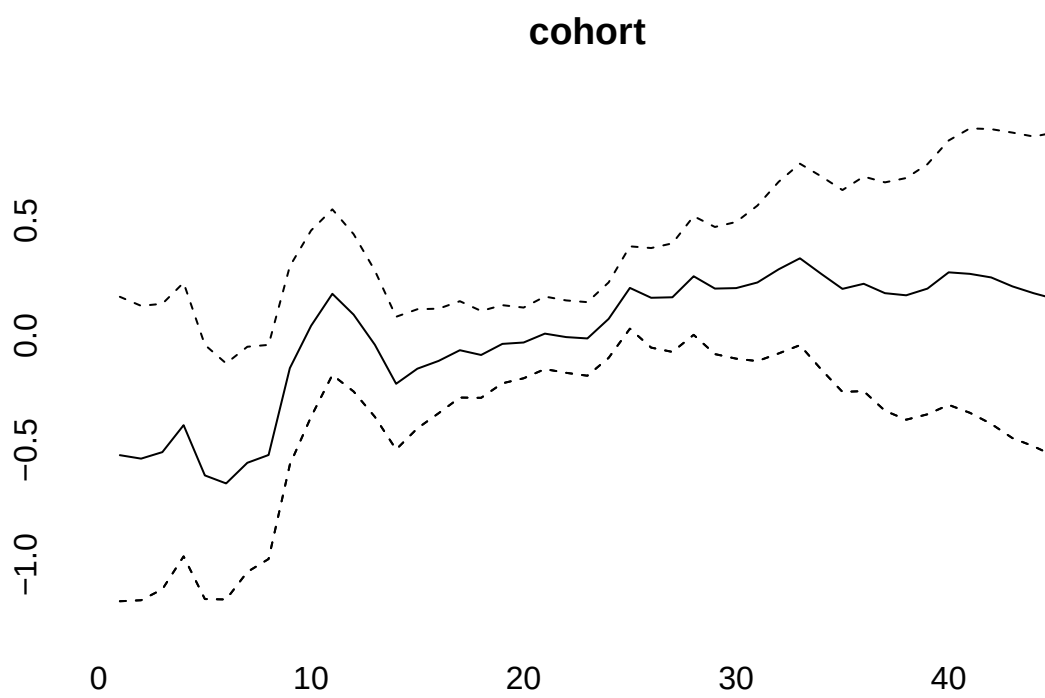
```
(model4<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
  cohort_covariate = cov_c, periods_per_agegroup = 5))
```

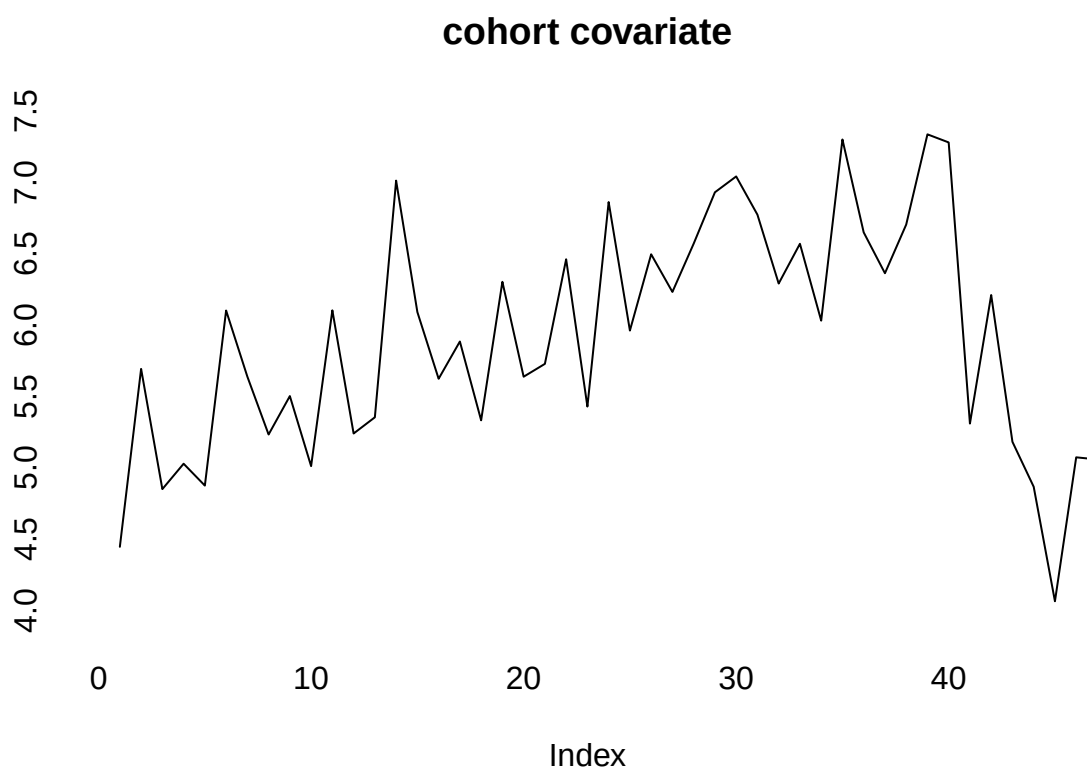
```
##
## Model:
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      231.66
## pD:            37.20
## DIC:           268.87
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.356         0.901         1.915
## period                     68.280        199.818        629.707
## cohort                     34.413         59.189         98.808
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model4)
```



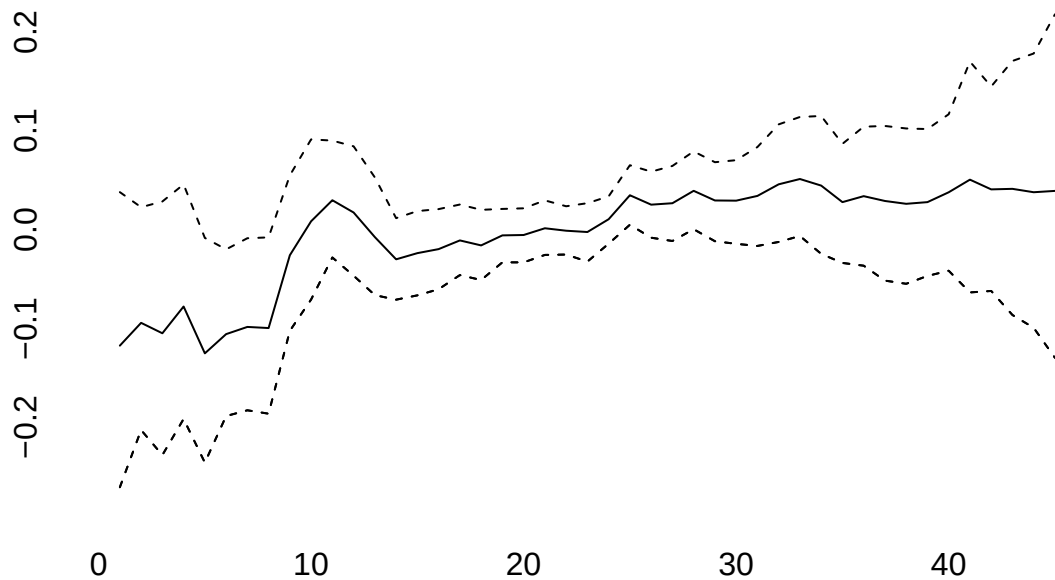








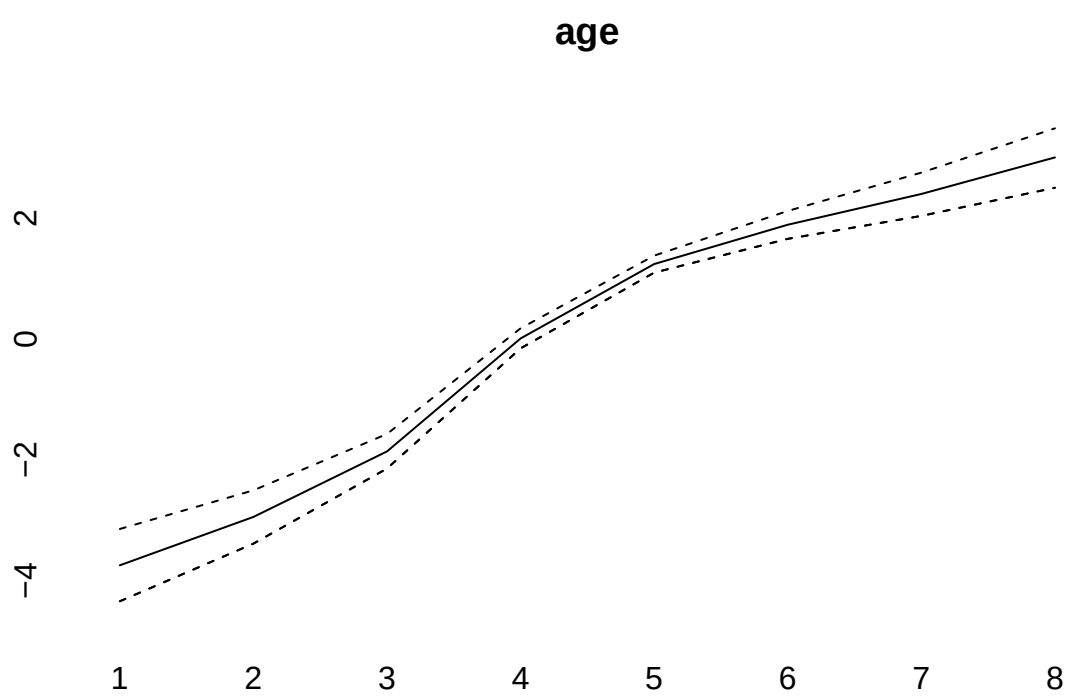
raw effect of cohort covariate

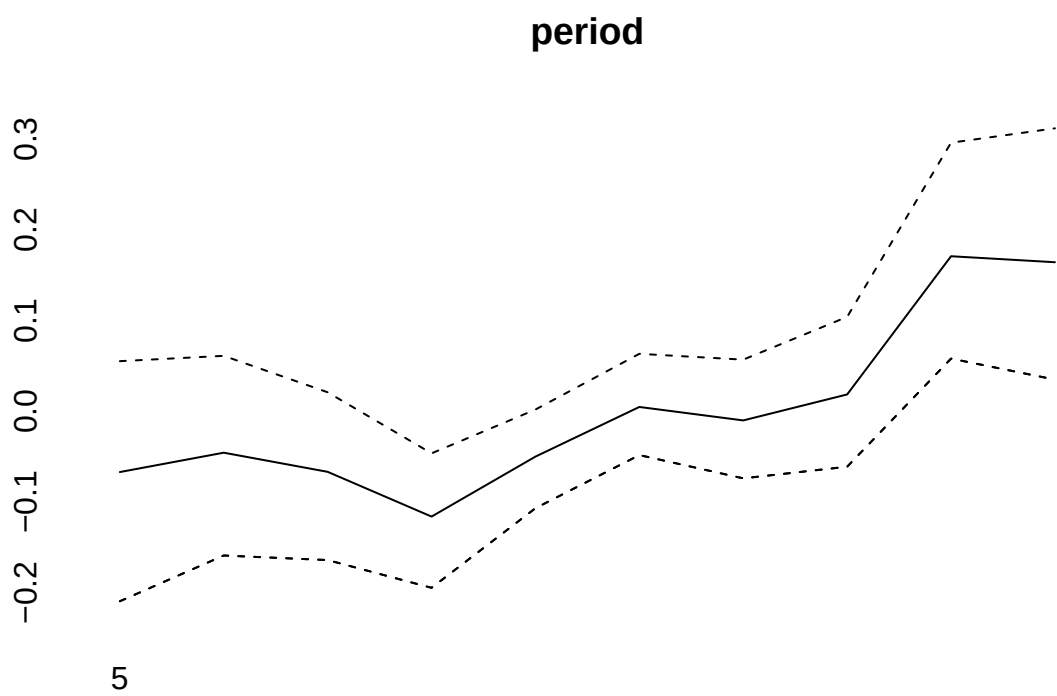


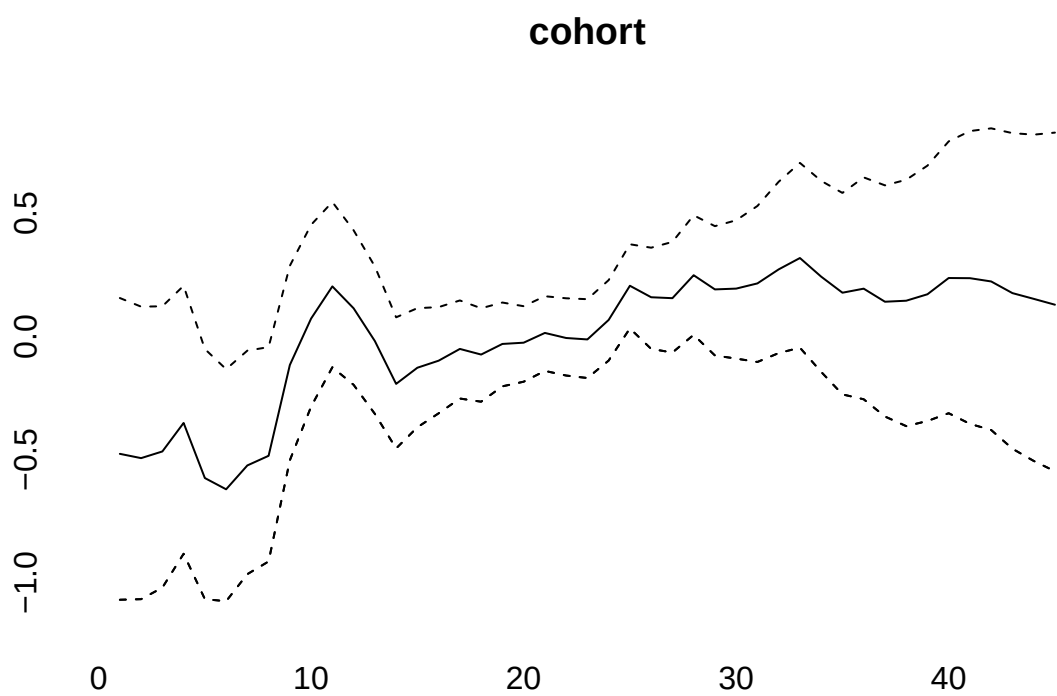
```
(model5<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
  period_covariate = cov_p, periods_per_agegroup = 5))
```

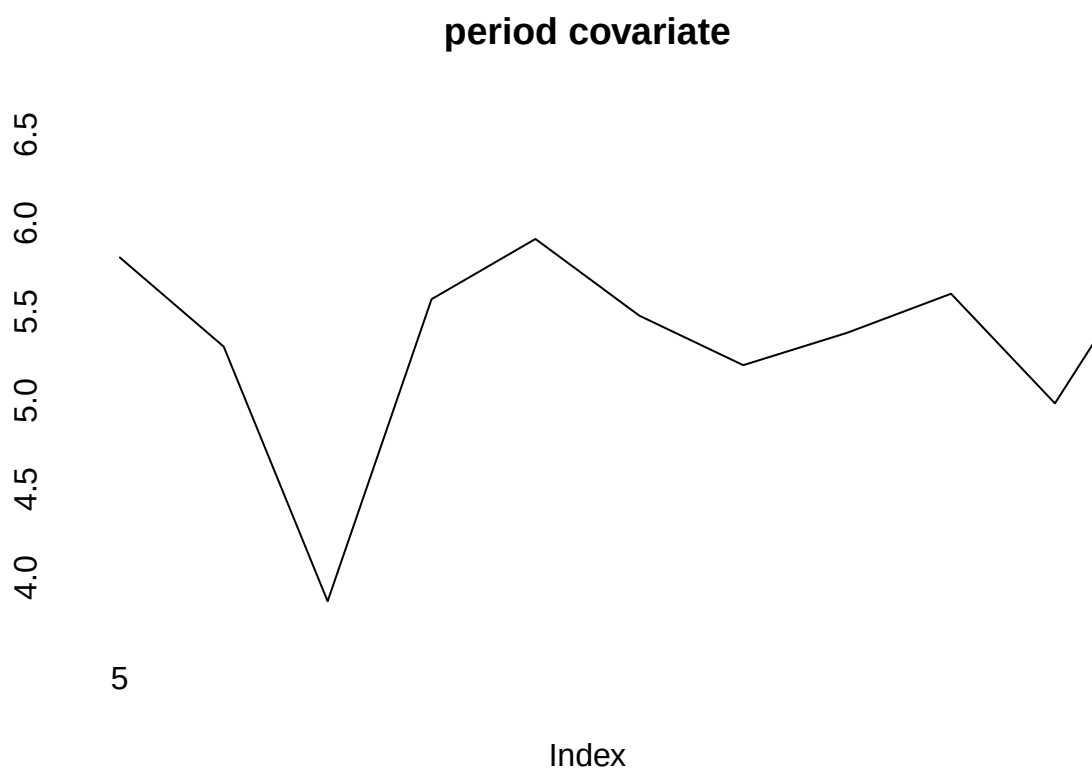
```
##
## Model:
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      231.63
## pD:            37.28
## DIC:           268.90
##
##
## Hyper parameters:           5%           50%           95%
## age                        0.358        0.934        1.997
## period                     66.952       193.989       581.799
## cohort                     35.324        59.802        97.410
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

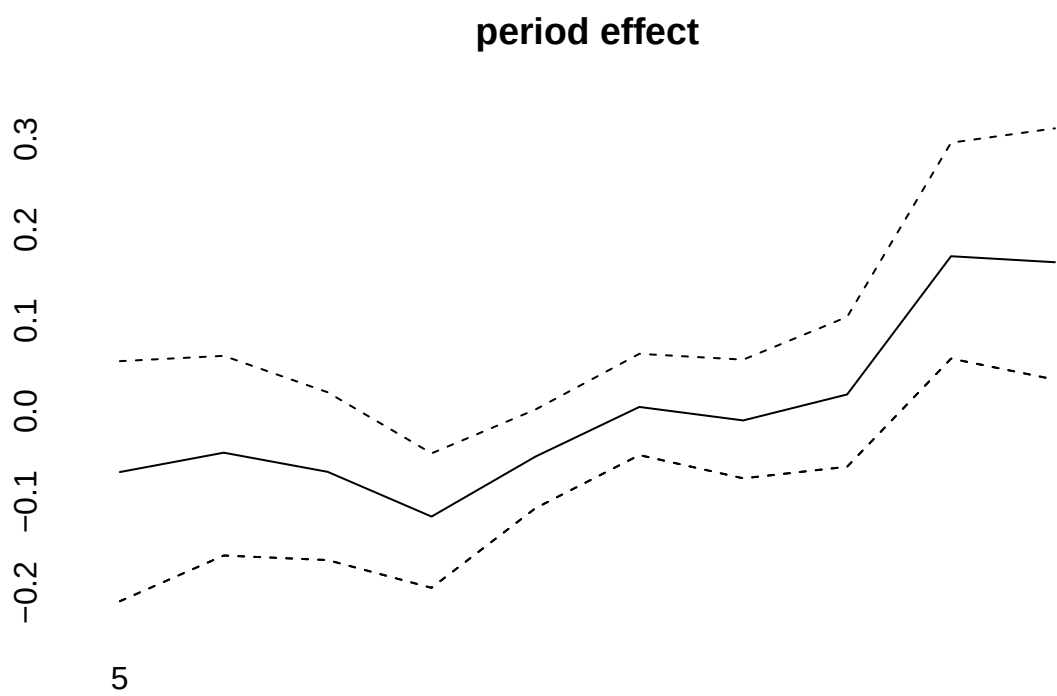
```
plot(model5)
```











raw period covariate effect

